Physical Energy Certificates for a Conditional Four-Dimensional Einstein Gap

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We study the normalization and high-curvature-tail interfaces in *An L^2 Einstein Gap for Nonnegatively Curved Four-Manifolds*. The source's topological conclusion remains conditional on its exact Einstein classification premise and on its geometric smoothing argument. At the normalization interface we retain the classical almost-Schur defect in a square identity valid for every trial scalar potential, including an inexact one. The full Poisson residual enters with its squared physical L^2 norm, with no inverse-Laplacian constant. An orthogonal augmented-Hessian representation explains this cancellation, and interval bounds turn it into a certificate for the normalized Ricci error and the initial high-curvature tail. We also give an explicit existing-time tail bootstrap, retaining its moving threshold rather than replacing it by an unspecified exponential constant. Round-sphere and product-sphere examples distinguish a full residual from a Galerkin residual and a topology statement from metric rigidity. No universal numerical topology threshold or independent validation of the classification theorem is obtained. Contribution assessment: C3/5, the inverse-free inexact energy identity and its retained-tail bootstrap give a substantive source-specific quantitative certificate, while the global topology theorem remains imported and conditional.

1 The Problem and Its Conditional Boundary

Let (M, g) be a smooth connected closed Riemannian four-manifold. Closed means compact without boundary. Write

$$R = \mathrm{Scal}_g := \mathrm{tr}_g \,\mathrm{Ric}_g, \quad V = \mathrm{Vol}_g(M), \quad \bar{R} = V^{-1} \int_M R \, d\mu_g, \quad E = \mathrm{Ric} - \frac{R}{4} g, \quad D = \int_M |E|_g^2 \, d\mu_g.$$

Here the notation Scal_g is only a name for the scalar R. All tensor contractions use g, and all integrals in the theorems are *unnormalized* volume integrals. Our Laplacian is $\Delta = \mathrm{tr}_g \,\mathrm{Hess}$, so $-\Delta$ is nonnegative. For $c > 0$, the metric cg has scalar curvature R/c and volume form $c^2 d\mu_g$; Ric and E as covariant two-tensors are unchanged. Thus $D(cg) = D(g)$. Dividing this energy by volume would destroy the scale invariance used in the source.

The assigned source [1] asserts the following implication. Assume that every connected smooth closed four-manifold satisfying $\mathrm{Ric} = 3g$ and $\sec \geq 0$ has universal Riemannian cover equal to the round S^4, suitably normalized Fubini–Study $\mathbb{CP}^2$, or the equal-radius product $S^2 \times S^2$. Then there exists a universal $\varepsilon_0 > 0$ such that a simply connected closed four-manifold with $\sec \geq 0$ and $D < \varepsilon_0$

is diffeomorphic to one of these three standard smooth manifolds, without a prescribed orientation. This is the source's `ass:classification` and `thm:gap`; it is not an unconditional conclusion of this report.

The hypothesis is an exact-Einstein classification, not a near-Einstein stability theorem. Its companion zero-plane proof is not independently audited here. Nor do we identify a numerical ε_0. The source does not require an auxiliary volume, diameter, injectivity, curvature-maximum, or Sobolev bound. Such bounds arise inside its contradiction argument; they must not be silently supplied by a finite calculation.

1.1 Where the local work enters

For a hypothetical sequence with $D_j \to 0$, the source chooses $\bar{R} = 12$, obtains $\| \operatorname{Ric} - 3g\|_2 \to 0$, and proves volume, diameter, and Sobolev bounds. It then smooths by Ricci flow using a small high-curvature part. Positive-time compactness produces an exactly Einstein limit. Nonnegative sectional curvature is recovered in that limit by an integral defect estimate and a second, small-time limit, not by assuming it persists along the flow.

Our principal change is at the first and second of these interfaces. The source's valid bound $\|R - 12\|_2 \le 4\|E\|_2$ is deliberately coarse. The classical sharp almost-Schur constant is $\sqrt{12}$ in dimension four [2]. We credit that improvement to the classical theorem. The contribution here is an exact *inexact-potential* account retaining the nonnegative defect and requiring the full continuum residual, followed by an explicit tail budget. Neither contribution replaces the geometric compactness theorem.

2 An Inexact Almost-Schur Square Identity

Put $U = R - \bar{R}$ and $S = \|U\|_2^2$. For a real $\psi \in H^2(M)$ define

$$H_\psi = \operatorname{Hess}\psi - \tfrac{1}{4}(\Delta\psi)g, \qquad r_\psi = \Delta\psi - U, \tag{2.1}$$

$$B_\psi = \int_M \operatorname{Ric}(\nabla\psi, \nabla\psi)\, d\mu_g, \qquad Q_\psi = 12\|E - H_\psi/3\|_2^2 + \tfrac{4}{3}B_\psi. \tag{2.2}$$

The smooth metric makes all these integrals meaningful for H^2 potentials. Constants in ψ do not matter. Ricci nonnegativity is not needed for the next identity, but is needed for $Q_\psi \ge 0$.

Theorem 2.1 (Full-residual square identity)**.** *On every smooth connected closed four-manifold,*

$$Q_\psi = 12D - S + \|r_\psi\|_2^2, \tag{2.3}$$

$$\left\|\operatorname{Ric} - \tfrac{\bar{R}}{4}g\right\|_2^2 = 4D - \tfrac{1}{4}Q_\psi + \tfrac{1}{4}\|r_\psi\|_2^2. \tag{2.4}$$

These formulas hold for every $\psi \in H^2(M)$; no Poisson equation, spectral lower bound, or finite-dimensional invariance is assumed.

Proof. First take ψ smooth. Contracted Bianchi gives $\operatorname{div} E = \frac{1}{4}dR$. Two integrations by parts, with no boundary term, give

$$\int_M \langle E, H_\psi\rangle \, d\mu_g = -\tfrac{1}{4}\int_M \langle dR, d\psi\rangle \, d\mu_g = \tfrac{1}{4}\int_M U\Delta\psi \, d\mu_g.$$

The integrated Bochner identity and the trace decomposition give

$$\|H_\psi\|_2^2 + B_\psi = \tfrac{3}{4}\|\Delta\psi\|_2^2. \tag{2.5}$$

Expanding the square in (2.2), its cross term is $-8\langle E, H_\psi\rangle_{L^2} = -2\langle U, \Delta\psi\rangle$. The remaining Hessian and Ricci-gradient terms sum to $\|\Delta\psi\|_2^2$ by (2.5). Hence

$$Q_\psi = 12D - 2\langle U, \Delta\psi\rangle + \|\Delta\psi\|_2^2 = 12D - S + \|\Delta\psi - U\|_2^2.$$

Since E is trace-free and $|g|_g^2 = 4$, $\|\operatorname{Ric} - \bar{R}g/4\|_2^2 = D + S/4$, proving the second identity. Smooth approximation in H^2, and boundedness of the smooth Ricci tensor on the compact manifold, extend both formulas to H^2. □

Corollary 2.2 (Exact potential and the classical constant)**.** *If* $\operatorname{Ric} \geq 0$*, let* f *be the mean-zero solution of* $\Delta f = U$*. Then*

$$S = 12D - 12\|E - H_f/3\|_2^2 - \tfrac{4}{3}B_f \leq 12D, \tag{2.6}$$

$$\|\operatorname{Ric} - \bar{R}g/4\|_2^2 = 4D - 3\|E - H_f/3\|_2^2 - \tfrac{1}{3}B_f \leq 4D. \tag{2.7}$$

Proof. The scalar Laplacian on a connected closed smooth manifold has constants as kernel and range equal to the mean-zero functions; $\int U = 0$ supplies compatibility. Its smooth mean-zero solution exists by elliptic theory. Apply Theorem 2.1 with $r_f = 0$ and $B_f \geq 0$. □

The factor $4D$ in (2.7) replaces the $5D$ obtainable from the source's coarse $S \leq 16D$. This is not a newly discovered sharp almost-Schur inequality. The useful retained quantity is $Q_f = 12D - S$, not merely the classical upper bound.

2.1 Why the residual is quadratic, without an inverse estimate

For $\operatorname{Ric} \geq 0$, use the real Hilbert space

$$\mathcal{H} = L^2(\operatorname{Sym}_0^2 T^*M, g) \oplus L^2(TM, g).$$

The square root of the nonnegative endomorphism $\operatorname{Ric}^\sharp$ exists pointwise, including at its zero eigenvalues. Define

$$Z_\psi = \left(\sqrt{12}(E - H_\psi/3), \tfrac{2}{\sqrt{3}}(\operatorname{Ric}^\sharp)^{1/2}\nabla\psi\right),$$
$$Tv = \left(-\tfrac{2}{\sqrt{3}}H_v, \tfrac{2}{\sqrt{3}}(\operatorname{Ric}^\sharp)^{1/2}\nabla v\right).$$

These are physical tensor and vector fields, not arrays endowed with an arbitrarily chosen Euclidean norm.

Proposition 2.3 (Augmented orthogonality)**.** *For* $v \in H^2(M)$*,*

$$\|Tv\|_{\mathcal{H}}^2 = \|\Delta v\|_2^2, \qquad Z_\psi = Z_f + T(\psi - f), \qquad \langle Z_f, Tv\rangle_{\mathcal{H}} = 0.$$

In particular, $Q_\psi = Q_f + \|r_\psi\|_2^2$*.*

Proof. The first equality is (2.5) multiplied by $4/3$. Polarization of that equality yields

$$\langle H_f, H_v\rangle + \int_M \operatorname{Ric}(\nabla f, \nabla v)\, d\mu_g = \tfrac{3}{4}\langle \Delta f, \Delta v\rangle.$$

The inner product of Z_f with Tv is therefore $-4\langle E, H_v\rangle + \langle U, \Delta v\rangle = 0$ by Bianchi. The affine decomposition follows from linearity of H and ∇. Pythagoras proves the last assertion. □

An inverse estimate for $\psi - f$ would generally need spectral information. The measured augmented quantity $T(\psi - f)$ does not: it has exactly the norm of the full Laplacian residual. This distinction is the reason an inverse constant disappears. It is not a claim that a poorly conditioned Poisson equation has become well conditioned in every norm.

3 Certified Normalization and Tail Budgets

Theorem 3.1 (Retaining a verified defect)**.** *Assume* $\mathrm{Ric} \geq 0$. *Suppose certified, unnormalized physical bounds give*

$$D \leq D_+, \qquad Q_\psi \geq Q_-, \qquad \|r_\psi\|_2^2 \leq e_+, \qquad Q_-, e_+, D_+ \geq 0.$$

Define

$$d_- = \max\{0, Q_- - e_+\}, \qquad S_+ = 12D_+ - d_-, \qquad L_+ = 4D_+ - d_-/4.$$

Then $0 \leq S \leq S_+$ *and* $\|\operatorname{Ric} - \bar{R}g/4\|_2^2 \leq L_+$. *If additionally* $\sec \geq 0$, $\bar{R} = 12$, *and* $|\operatorname{Rm}| \leq C_0 R$ *is a valid pointwise curvature estimate, put* $w_0 = (|\operatorname{Rm}| - 12C_0)_+$. *If a certified* $N_- \leq \int_M (U_-)^2 \, d\mu_g$ *is available, then*

$$\|w_0\|_2^2 \leq C_0^2 (S_+ - N_-). \tag{3.1}$$

The right sides are nonnegative whenever the stated enclosures are consistent. A negative right side is an enclosure failure, not a negative-energy prediction.

Proof. Proposition 2.3 gives $Q_f = Q_\psi - \|r_\psi\|_2^2 \geq d_-$. Use $S = 12D - Q_f$ and the trace decomposition. Pointwise, $w_0 \leq C_0 U_+$. Also $\|U_+\|_2^2 = S - \|U_-\|_2^2$, proving (3.1). □

For example, the abstract rational enclosure $D_+ = 1/100$, $Q_- = 1/20$, $e_+ = 1/1000$, $N_- = 1/100$ gives

$$d_- = 49/1000, \quad S_+ = 71/1000, \quad L_+ = 111/4000, \quad \|w_0\|_2^2 \leq (61/1000)C_0^2.$$

These numbers illustrate compatible budget arithmetic, not an assertion that a particular uncomputed metric realizes the enclosures. They improve respectively the source-derived bounds $S \leq 16D_+$ and $\|\operatorname{Ric} - 3g\|_2^2 \leq 5D_+$, and may also improve the classical bounds when $d_- > 0$.

3.1 Physical implementation and omitted components

Choose globally defined H^2 scalar trial functions ϕ_i and $\psi = \sum_i c_i \phi_i$. Local chart functions must be glued with their derivatives and connection terms; discontinuous panel functions are not automatically in this domain. The exact residual quadratic is

$$\|r_\psi\|_2^2 = c^T K c - 2c^T b + S, \quad K_{ij} = \int_M \Delta\phi_i \, \Delta\phi_j \, d\mu_g, \quad b_i = \int_M U \Delta\phi_i \, d\mu_g. \tag{3.2}$$

This formula is useful only if its integrals, including S, are bounded in the physical measure. Cancellation in the polynomial requires outward error bounds. When a direct interval integral of $|r_\psi|^2$ is cheaper, it avoids that cancellation. Neither procedure assumes a finite solve measures its own omitted residual.

The other square can be evaluated in a tensor bank with Gram entries $\int \langle H_{\phi_i}, H_{\phi_j} \rangle_g d\mu_g$ and Ricci-gradient entries $\int \mathrm{Ric}(\nabla\phi_i, \nabla\phi_j) d\mu_g$. Bianchi and (2.5) are compatibility checks between these banks, not licenses to erase unbounded quadrature errors. If a finite orthogonal measurement projection Π is used instead,

$$\|r_\psi\|_2^2 = \|\Pi r_\psi\|_2^2 + \|(I - \Pi) r_\psi\|_2^2.$$

The second term must have its own certificate. Positivity of a finite Gram matrix says nothing about it.

The construction does not make every term easier to compute than S directly. Its advantages are a second invariant consistency check, a reusable Poisson approximation, and a retained positive budget in a pipeline that already has these quantities. Its cost is reliable tensor quadrature and continuum residual control.

4 A Quantified Existing-Time Tail Bootstrap

This section makes a source propagation step explicit. It does not prove a common Ricci-flow lifespan. The distinction matters because the source establishes that lifespan only after noncollapse and a curvature-concentration contradiction.

Theorem 4.1 (Threshold-retaining propagation)**.** *Let $g(t)$ be a smooth Ricci flow on a closed four-manifold for $0 \le t < \mathcal{T}$, with $R \ge 0$. Suppose on the existing times $0 \le t < T$ that $u = |\operatorname{Rm}|$ satisfies $(\partial_t - \Delta)u \le qu^2$ weakly, $R \le cu$, and*

$$\|f\|_{4,t}^2 \le S_1 \int_M \{4|\nabla f|^2 + (R+1)f^2\}\, d\mu_t \tag{4.1}$$

for every Lipschitz f. Here q, c, S_1, K_0 are positive constants, $T < (qK_0)^{-1}$, and

$$K(t) = \frac{K_0}{1 - qK_0 t}, \quad w = (u - K)_+, \quad y(t) = \|w(t)\|_{2,t}.$$

Choose

$$0 < \eta < \frac{1}{S_1(c + 4q)}, \qquad d = cS_1\eta < 1. \tag{4.2}$$

Set

$$F_\eta(t) = (1 - qK_0 t)^{-(2-d)/(1-d)} \exp\!\left(\frac{qS_1\eta t}{1-d}\right). \tag{4.3}$$

If $y(0)F_\eta(T) < \eta$, then, on every existing time $t < \min(T, \mathcal{T})$,

$$y(t) \le y(0)F_\eta(t) < \eta, \tag{4.4}$$

$$\|f\|_{4,t}^2 \le \frac{S_1}{1-d} \int_M \{4|\nabla f|^2 + (cK(t) + 1)f^2\}\, d\mu_t. \tag{4.5}$$

Proof. On an interval with $y \le \eta$, use $R \le c(K + w)$ and Hölder to move $cS_1 \int wf^2 \le d\|f\|_4^2$ to the left of (4.1). This proves (4.5), with its stated two coefficients. The threshold satisfies $K' = qK^2$; the positive-part energy calculation from the source gives, writing $Y = y^2$ and $G = \int |\nabla w|^2 d\mu_t$,

$$Y' \le -2G + 2q \int w^3 d\mu_t + 4qKY.$$

The volume derivative is $-R\, d\mu_t$ and has favorable sign. The usual Lipschitz positive-part approximation justifies this calculation, including the set where $u = K > 0$. Combining Hölder with (4.5),

$$Y' \le \left(-2 + \frac{8qS_1\eta}{1-d}\right) G + \left(4qK + \frac{2qS_1\eta(cK+1)}{1-d}\right) Y.$$

The gradient coefficient is negative by (4.2). Dropping it and integrating the resulting linear inequality for Y gives the square of (4.3); this also handles $Y(0) = 0$. Indeed, the coefficient for y is

$$\frac{q(2-d)}{1-d} K(t) + \frac{qS_1\eta}{1-d}, \qquad q \int_0^t K(s)\, ds = -\log(1 - qK_0 t).$$

The strict hypothesis at T, continuity, and a first-crossing argument rule out $y = \eta$ before $\min(T, \mathcal{T})$. □

For the source one takes $K_0 = 12C_0$. Its weighted Sobolev estimate, curvature reaction constant, and scalar-to-curvature norm comparison must supply S_1, q, c. The theorem does not manufacture these constants from a mesh. The output of Theorem 3.1 may supply $y(0)$, with its original physical scale and volume. This replaces an unspecified smallness choice and unspecified e^{Ct} by a checkable sufficient condition. It is not an optimal threshold and does not improve the exponent in any smoothing theorem.

Example 4.2 (A rational bootstrap budget)**.** As abstract supplied constants, take $q = S_1 = K_0 = 1$, $c = 3$, $\eta = 1/16$, $T = 1/10$, and $y(0) \le 1/32$. The absorption condition is $1/16 < 1/7$. Here $d = 3/16$ and

$$F_\eta(T) = (10/9)^{29/13} e^{1/130} < (10/9)^3 \frac{130}{129} = \frac{130000}{94041} < 2.$$

The exponential bound follows from $e^x \le (1-x)^{-1}$ for $0 \le x < 1$, by the power series. Thus the strict guard holds. This is an exact arithmetic test of the criterion, not a numerical claim that those constants have been proved for a particular nonnegatively curved manifold.

5 Geometric Fixtures and Failure Modes

5.1 An unequal product, with an inexact potential

Let $M = S^2(k_1) \times S^2(k_2)$, where $S^2(k)$ is the round two-sphere of sectional curvature $k > 0$. In a product orthonormal frame,

$$V = \frac{16\pi^2}{k_1 k_2}, \qquad R = 2(k_1 + k_2), \qquad D = V(k_1 - k_2)^2, \qquad S = 0. \tag{5.1}$$

The eigenvalues of E are $(k_1 - k_2)/2$ twice and its negative twice. Taking $\psi = A\cos\theta_1$, set $m = A^2 k_1^2 V/3$. On the first factor $\operatorname{Hess}\psi = -k_1\psi g_1$, while it vanishes on the second. Since the spherical average of $\cos\theta_1$ is zero and that of $\cos^2\theta_1$ is $1/3$,

$$\|H_\psi\|_2^2 = m, \quad B_\psi = 2m, \quad \|r_\psi\|_2^2 = 4m, \quad Q_\psi = 12D + 4m. \tag{5.2}$$

In particular, the residual term cannot be omitted even though the underlying scalar curvature is already constant. The arbitrary trial potential is not the exact mean-zero Poisson solution. With $\psi = 0$, the certificate instead reads $Q = 12D$ and recovers $S = 0$ and $\|\operatorname{Ric} - \bar{R}g/4\|_2^2 = D$ exactly.

After $k_1 + k_2 = 6$, the source normalization is already imposed. Writing $k_1 = 3 + t$, $k_2 = 3 - t$, $|t| < 3$, gives

$$D = \frac{64\pi^2 t^2}{9 - t^2}.$$

More invariantly, if $z = \frac{1}{2}\log(k_1/k_2)$, then $D = 64\pi^2 \sinh^2 z$. Thus this particular family has the exact aspect-ratio control $|z| \le \operatorname{arsinh}(\sqrt{D}/(8\pi))$. It is a family-specific calculation, not a distance-to-Einstein estimate for arbitrary four-manifolds. All these unequal products have an allowed topology but are not Einstein.

5.2 A projected residual can give a false energy inequality

On the unit round S^4, $E = 0$, $U = 0$, and $\operatorname{Ric} = 3g$. Take $\psi = x_1 x_2$, the restriction of a harmonic quadratic polynomial in $\mathbb{R}^5$. Then $\Delta\psi = -10\psi$, and, with $V = \operatorname{Vol}(S^4)$,

$$\int \psi^2 \, d\mu_g = \frac{V}{35}, \quad B_\psi = \frac{30V}{35}, \quad \|H_\psi\|_2^2 = \frac{45V}{35}, \quad Q_\psi = \|r_\psi\|_2^2 = \frac{100V}{35}.$$

These also verify the coefficient one on the residual square in (2.3). It cannot be uniformly reduced while leaving the other terms unchanged. The measurement projection onto constants has $\Pi r_\psi = 0$, but $r_\psi \neq 0$. Replacing its full norm by that projection would falsely imply $S < 0$ from (2.3). This counterexample has the very same positive curvature and closed domain as the intended application; it is not an unrelated matrix model.

5.3 Collapse without the topological hypotheses

Take $S^2(6) \times T^2_a$, with a flat two-torus of area a. Its sectional curvature is nonnegative and $\bar{R} = 12$. Its Ricci eigenvalues are $6, 6, 0, 0$, hence

$$|E|^2 = 36, \quad V = (2\pi/3)a, \quad D = 24\pi a \longrightarrow 0.$$

Its fundamental group is $\mathbb{Z}^2$, and it is not one of the three simply connected model manifolds. This does not contradict the source. It proves that small scale-invariant D and the displayed scalar normalization alone do not prevent collapse when its simple-connectivity/topological input is discarded. The original proof uses $\chi = 2 + b_2 \geq 2$ precisely before establishing a lower volume bound.

5.4 Coordinate error is not curvature error

For a covariant symmetric tensor T in a chart with metric matrix G, the physical integrand is

$$|T|^2_g d\mu_g = \operatorname{tr}(G^{-1}TG^{-1}T)\sqrt{\det G}\, dx.$$

A change $x = A\xi$ transforms G and T to A^TGA and A^TTA; cyclicity of trace preserves the contraction, and the measure acquires $|\det A|$. In the special flat chart $G = I$, $T = \operatorname{diag}(1, -1, 0, 0)$, and $A = \operatorname{diag}(s, s^{-1}, 1, 1)$, the determinant is one but the naive sum of squared transformed components is $s^4 + s^{-4}$, not the physical value 2. This tensor test is not a realization of an arbitrary tensor as the trace-free Ricci tensor. It verifies the norm rule needed when actual curvature components are supplied.

6 The Source Route and Its Unreplaced Inputs

The entire assigned mathematical source is one 863-line file. The following map refers to its literal line numbers, not to the numbering of this report.

Source lines	Mechanism	Boundary retained here
56–190	Classification premise and conditional gap	Companion exact classification remains an assumption.
192–297	Scale, almost-Schur, Euler energy, segment diameter	Local square refinement; global geometric estimates not replaced.
299–394	Support truncation and initial Sobolev inequality	Actual geodesic kernel, physical measure, and lower volume needed.
396–480	Entropy to weighted Sobolev at existing times	Perelman input and uniform entropy constants are imported.
482–568	Moving high-curvature threshold	Explicit bootstrap in Theorem 4.1 under stated inputs.
570–646	Noncollapse, Shi estimate, common lifespan	Small full tail and all-space Sobolev required, not sample bounds.
648–752	Einstein-error and negative-sectional-defect propagation	Moving frames and the integral Ct bound retained.
754–859	Positive-time compactness, shrinking limit, classification	Compactness, diffeomorphism recovery, and both limits remain global.

For clarity, three delicate source details are preserved. First, the weighted Sobolev bound is proved before a common flow interval is available. Its entropy parameter $\tau + t$ is not a future flow time. Our existing-time qualifier is consequently essential.

Second, the source obtains noncollapse by iterating a cutoff inequality on one fixed smooth slice. The Euclidean small-ball limit is taken on that slice; no uniform initial injectivity scale is presumed. Its subsequent first-crossing argument rescales a curvature maximum, applies a derivative estimate, and turns that maximum into a fixed positive L^2 tail mass. Neither a finite spectral solve nor a collection of low residual samples supplies that contradiction.

Third, nonnegative sectional curvature is not assumed preserved under Ricci flow. Let $F(\mathrm{Rm})$ denote the negative part of the minimum sectional curvature. The source proves $\int F(\mathrm{Rm}(g_j(t)))d\mu_{g_j(t)} \leq Ct$. After smooth positive-time convergence to $(1-6t)h$, this becomes

$$(1-6t)\int F(\mathrm{Rm}(h))d\mu_h \leq Ct.$$

Only now may $t \downarrow 0$ recover $\mathrm{sec}_h \geq 0$. That order of limits does not assert smooth convergence of the original metrics at time zero.

7 Ram Proof Connections and Scope

The original physical-metric chapter [5], lines 1–210, derives $G = J^*J$, projection $JG^{-1}J^*$, the orthogonal full-residual decomposition, and a relative metric error rule. It explicitly separates a compression from an invariant space and a finite measurement bank from a continuum frame. Here those proofs apply to the actual Hilbert sum in Proposition 2.3, not to a replacement norm. The special gain is the geometric isometry $\|Tv\|_{\mathcal{H}} = \|\Delta v\|_2$ derived from Bianchi–Bochner. It is not supplied by the general Gram identity alone.

Fresh original coordinate proofs [6] were read at lines 1–140, 262–282, 334–350, and 392–426. They derive the measure-correct pullback, divergence-form Laplacian, nonlinear chain rule, and covariant Hessian. The two included files named at lines 65 and 117 were not read and are not cited as proved dependencies. We use the displayed body proofs. For this report they require $(\mathrm{Hess}\,\psi)_{ij} = \partial_i\partial_j\psi - \Gamma^k_{ij}\partial_k\psi$, rather than ordinary second partial derivatives in a nonlinear chart.

The original geometry and energy chapter [7], lines 942–1255, derives the two-point frame rule, connection transformation, transported Taylor remainder, curvature commutator, and compact-support energy identity with its boundary-form qualification. These are genuinely geometric operator proofs, but they are not an Einstein classification theorem. Their local transport formula requires a normal neighborhood and a metric-compatible connection. They do not supply a uniform injectivity radius for the source sequence.

The scope is 737 freshly displayed private lines in these three original files. Unread complements, all included-file exclusions, and PRE/POST hashes are recorded in the read ledger. No exhaustive search or proof audit of the confidential corpus is claimed. The comparison is a synthesis of these proved metric principles with the specific curvature identities above, not a claim of historical novelty for Bianchi, Bochner, or almost-Schur.

8 Status, Advantages, and Further Work

The public primary refresh on October 9, 2026 checked the assigned repository README and date, the De Lellis–Topping arXiv version record and primary proof, Liu's published toric classification statement, and the Chan–Chen–Lee publication abstract and stated smoothing context. The exact scopes matter. The almost-Schur constant and its Ricci-error corollary are classical [2]. Liu's theorem has a T^2 symmetry hypothesis [3]; it does not establish the source's unrestricted classification assumption. Chan–Chen–Lee impose geometric concentration and noncollapse/entropy conditions [4]; we did not independently audit their entire proof. Public availability of the assigned preprint is not independent acceptance of its global claim.

The local advantages are concrete: an arbitrary admissible potential has a quadratic full-residual account; no unknown Poisson inverse constant is needed for this particular augmented norm; retained deficits can reduce the initial high-curvature-tail budget; and the bootstrap displays exactly where absorption and the moving threshold restrict a time horizon. The costs are equally concrete: full physical residual integration, connection and measure control, lower bounds for a positive square, and supplied weighted Sobolev and curvature constants. Computing these quantities may be harder than directly bounding scalar variance. No runtime improvement is asserted.

Natural next steps are rigorously validated atlas quadrature for an explicit metric, residual estimators compatible with the actual H^2 domain, and numerical optimization of η under already proved flow constants. Such optimization must not conceal S_1 or the dependence of K_0 on normalization. Extending the identity to manifolds with boundary requires the actual Bianchi and Bochner boundary terms and boundary conditions; merely using a finite Dirichlet matrix is insufficient. A theorem converting small D into distance from a model Einstein metric would require additional gauge, tail, and global stability analysis not provided here.

9 Verification and Conclusion

The owned executable checks test the square algebra, actual product-sphere integral fixtures, sphere harmonic residuals, coordinate-contraction invariance and its Euclidean-norm negative control, and rational absorption and bootstrap bounds. They pass 108 exact parameter or symbolic cases comprising 758 assertions; there are no floating-point diagnostic cases. The final test receipt distinguishes parameter cases from asserted identities and is tied to the executed script hash. It contains no parent or foreign-checker counts. Finite fixtures supplement the displayed proofs; they do not validate Ricci-flow compactness, classification, or a universal topology threshold.

All 863 lines of the assigned scientific manuscript, 191 bibliography lines, and 18 README lines were freshly displayed in bounded untruncated intervals. The full source architecture was inspected; external global theorems cited by it were not thereby independently proved. The final read ledger retains that distinction. No source error is alleged in the deliberately coarse factor 4: it is a valid estimate with unused trace-free information.

The conclusion is a sharper, residual-aware normalization and tail interface for the source's conditional gap argument. The physical energy is never replaced by a finite coefficient norm. The topological conclusion, its noncollapse mechanism, and its classification assumption remain exactly where the source puts them.

References

[1] OpenAI, *An L^2 Einstein Gap for Nonnegatively Curved Four-Manifolds*, October 5, 2026. Assigned pinned source, `build/source/main.tex`, lines 1–863. https://github.com/openai/math/tree/main/preprints/An-L2-Einstein-Gap-for-Nonnegatively-Curved-Four-Manifolds-October-5-2026. Repository metadata checked; the conditional global theorem is not independently certified by this report.

[2] C. De Lellis and P. M. Topping, *Almost-Schur lemma*, Calc. Var. PDE **43** (2012), 347–354, https://doi.org/10.1007/s00526-011-0413-z. arXiv:1003.3527v2, May 7, 2011. Theorem 1.1, Corollary 1.2, and Section 2.1 read in the author's primary PDF.

[3] T. Liu, *Toric Einstein 4-manifolds with non-negative sectional curvature*, Ann. Global Anal. Geom. **67** (2025), article 14, published April 2, 2025. https://doi.org/10.1007/s10455-025-09990-3. Theorem 1.1 and publication metadata read; full proof not audited.

[4] P.-Y. Chan, E. Chen and M.-C. Lee, *Small curvature concentration and Ricci flow smoothing*, J. Funct. Anal. **282** (2022), 109420, https://doi.org/10.1016/j.jfa.2022.109420. Primary abstract and introductory hypotheses inspected; external smoothing machinery remains an input.

[5] M. Subbarao, *RamCalBooks*, Volume 21, `chapters/ch01d-physical-metrics.tex`, private original, lines 1–210; especially `v21c01:residual-decomposition` and `v21c01:residual-identity`. Local confidential reading only.

[6] M. Subbarao, *RamCalBooks*, Volume 1, revision 2, `chapters/ch06-coordinates-rev2.tex`, private original, lines 1–140, 262–282, 334–350, 392–426. Included files are outside the proof-read scope.

[7] M. Subbarao, *RamCalBooks*, Volume 2, `chapters/ch10-hypercomplex-geometry.tex`, private original, lines 942–1255; geometric connection and covariant-moment proofs, and energy/boundary qualifications. Other chapter sections unread.